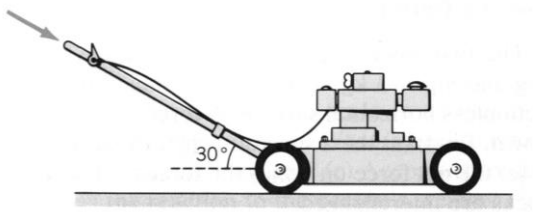
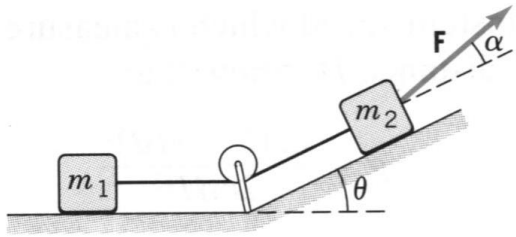
Homework #7

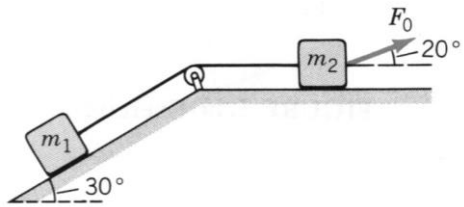
1) A man pushes a 20.0 kg lawn mower with a force of 80.0 N directed along the handle, which is inclined at 30.0° to the horizontal as shown in the figure below. (a) If he moves at constant velocity, what is the impeding force due to the ground? (b) What force along the handle would produce an acceleration of 1.25 m/s^2 given the same impeding force?



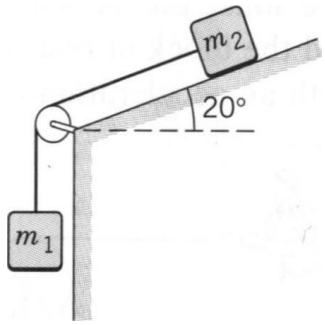
2) Two blocks, $m_1 = 2.00$ kg and $m_2 = 4.00$ kg, are connected via a rope that passes under a pulley as in the figure below. The block of mass is on a frictionless surface inclined at $\theta = 25°$ to the horizontal and is subject to a force F = 26.0 N that is directed at $\alpha = 15°$ to the slope. Determine the acceleration of the blocks and the tension in the rope.



3) Two blocks with masses $m_1 = 3.00$ kg and $m_2 = 5.00$ kg are connected by a light rope and slide on a frictionless surface as the figure below. A force $F_0 = 10.0$ N acts on m_2 at 20° to the horizontal. Find the acceleration of the system and the tension in the rope.



4) A block of mass $m_1 = 1.15$ kg is connected by a rope to a block of mass $m_2 = 2.00$ kg. The latter is on a frictionless 20° incline as shown in the figure below. Find the acceleration of the blocks and the tension in the rope.



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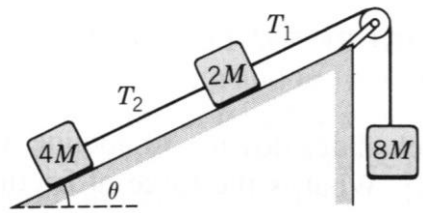
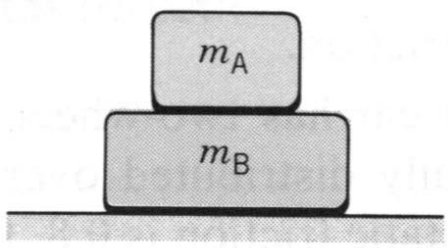
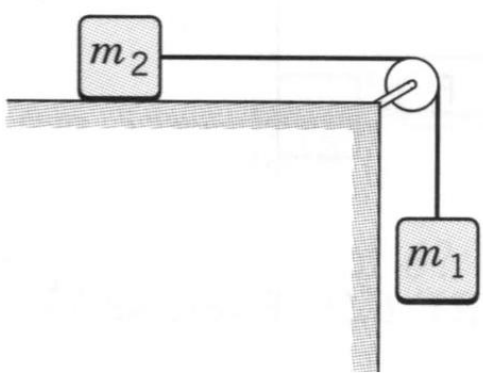
5) Three blocks with masses $4M$, $2M$, and $8M$ are connected as shown in the figure below. The tensions in the ropes are T_1 and T_2. In terms of M, g, and θ, obtain expressions for: (a) the acceleration; (b) $T_1 - T_2$. Ignore friction.



6) A 2.50-kg block is on a 53.0° incline for which $\mu_k = 0.25$ and $\mu_s = 0.50$. Find its acceleration given that: (a) it is initially at rest; (b) it is moving up the slope; (c) it is moving down the slope.

7) Block A of mass $m_A = 2.00$ kg is on block B of $m_B = 3.00$ kg as shown in the figure. The lower block is on a frictionless surface while $\mu_s = 0.250$ between the two blocks. (a) If they are moving at constant velocity, what is the frictional force between A and B? (b) What is the maximum horizontal force that can be applied to B without A slipping?



8) Two blocks with masses $m_1 = 0.820$ kg and $m_2 = 1.40$ kg are connected by a string as shown in the figure. If $\mu_k = 0.200$ for the horizontal surface, find the acceleration.